That is the corresponding angles are equal.

Therfore, l_1 and l_2 are parallel.

Thus, non-vertical lines having equal slopes are parallel.

Hence, non vertical lines are parallel if and only if their slopes are equal.

5.4.3 Slopes of perpendicular lines

Two non-vertical lines with slopes m_1 and m_2 are perpendicular if and only if $m_1 m_2 = -1$.

Let l_1 and l_2 be two non-vertical lines with slopes m_1 and m_2, respectively. Let their inclinations be θ_1 and θ_2 respectively.

Then $m_1 = \tan\theta_1$ and $m_2 = \tan\theta_2$

First we assume that, l_1 and l_2 are perpendicular to each other.

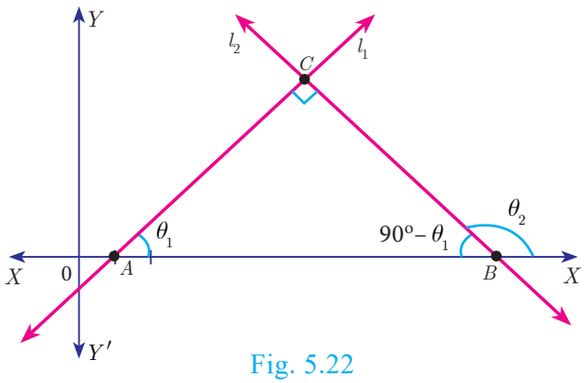


Fig. 5.22

Then $\angle ABC = 90^\circ - \theta_1$ (sum of angles of ΔABC is 180°)

Now measuring slope of l_2 through angles θ_2 and $90^\circ - \theta_1$, which are opposite to each other, we get

$$\tan\theta_2 = -\tan(90^\circ - \theta_1)$$

$$= \frac{-\sin(90^\circ - \theta_1)}{\cos(90^\circ - \theta_1)} = \frac{-\cos\theta_1}{\sin\theta_1} = -\cot\theta_1 \quad \text{gives, } \tan\theta_2 = -\frac{1}{\tan\theta_1}$$

$$\tan\theta_1 . \tan\theta_2 = -1$$

$$m_1 . m_2 = -1 .$$

Thus, when the line l_1 is perpendicular to line l_2 then $m_1 m_2 = -1$.

Conversely,

Let l_1 and l_2 be two non-vertical lines with slopes m_1 and m_2 respectively, such that $m_1 m_2 = -1$.

Since $m_1 = \tan\theta_1$, $m_2 = \tan\theta_2$

We have $\tan\theta_1 \tan\theta_2 = -1$

$$\tan\theta_1 = -\frac{1}{\tan\theta_2}$$

$$\tan\theta_1 = -\cot\theta_2$$

$$\tan\theta_1 = -\tan(90^\circ - \theta_2)$$

$$\tan\theta_1 = \tan\left(-(90^\circ - \theta_2)\right) = \tan(\theta_2 - 90^\circ)$$

$$\theta_1 = \theta_2 - 90^\circ \qquad (\text{since } 0 \le \theta_1 \le 180^\circ,\ 0 \le \theta_2 \le 180^\circ)$$

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